A Configuration-Component-Based Hybrid Model for Combined-Cycle Units in MISO Day-Ahead Market

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***Abstract*—This paper proposes a hybrid combined-cycle gas turbine (CCGT) model for day-ahead market clearing, in order to enhance the operation flexibility of CCGTs in practice. The proposed hybrid model, by taking benefits of combined offers from market participants on both configurations and individual physical turbines, can more accurately reflect physical operation features of CCGTs than existing CCGT models. A comprehensive review on existing CCGT models in academia and industry practice with their advantages and shortcomings is conducted. By taking benefits of the two most investigated models, i.e., configuration-based model and component-based model, the mapping relationship between these two models is revealed for deriving the proposed hybrid model. Tight formulations are further discussed for achieving the better computational performance. The proposed hybrid model is tested and compared with other CCGT models via the modified IEEE 118-bus system and the midcontinent independent system operator system. Results show notable benefits in maintaining operation flexibility and enhancing social welfare.**

***Index Terms*—Combined-cycle gas turbine, hybrid model, mixed-integer programming, security-constrained unit commitment, tight formulation.**

NOMENCLATURE

Major symbols and notations used throughout the paper are listed below, while others are defined following their first appearances as needed.

Acronyms

CAISO	California Independent System Operator
CCGT	Combined-cycle gas turbine
CFBM	Configuration-based model
CPBM	Component-based model
CT/ST/DB	Combustion turbine/steam turbine/duct burner
EBM	Edge-based model
ERCOT	Electric Reliability Council of Texas
HRSG	Heat recovery steam generator
ISO-NE	Independent System Operator New England
LP	Linear programming
MIP	Mixed-integer programming
MISO	Midcontinent Independent System Operator
MP	Market participant
NERC	North American Electric Reliability Corporation
PT	Physical turbine, including CT and ST
SCUC	Security-constrained unit commitment
SPP	Southwest Power Pool

Indices

c, s	Indices of CTs and STs per CCGT, i.e., $c, s \in X$
j	Index of CCGTs, i.e., $j \in J$
t, t'	Indices of time intervals (hours), i.e., $t, t' \in T$
w	Type of PT startup warmth status, i.e., $w \in W$
x	Index of PTs per CCGT, $x \in X$
$y,\ y'$	Indices of configurations per CCGT, i.e., $y,\ y' \in Y$

Sets

$F_j^{yy'}$	Set of feasible transitions of CCGT j, where y/y' is the "from"/"to" configuration and $y \neq y'$
$J/\ T$	Set of CCGTs/time intervals
ON_j^x	Set of valid configurations with PT x on for CCGT j
UT_j^x/DT_j^x	Set of upward/downward transitions which involve the startup/shutdown of PT x for CCGT j
W_j	Set of startup warmth status types for CCGT j, i.e., $W_j = \{1 = \text{hot}; 2 = \text{intermediate}; 3 = \text{cold}\}$
$X_j/\ Y_j$	Set of PTs/valid configurations of CCGT j

Variables

$i_{j,t}^x$	Intermediate variable for commitment status of PT x for CCGT j at time t
$p_{j,t}^y/u_{j,t}^y$	Power output/commitment status of CCGT j at configuration y and time t
$su_{j,t}^x/sd_{j,t}^x$	Intermediate variable for startup/ shutdown indicators of PT x for CCGT j at time t

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$v_{j,t}^{yy'}$	Indicator of transition from y to y' ($y \neq y'$) for CCGT j at time t
$\delta_{j,t,w}^{x}$	Indicator of startup type w of PT x for CCGT j at time t

Constants

$C_{j,w}^{SU,x}$	Startup cost of PT x for CCGT j with type w
$C_{j}^{P,y}(\cdot)$	Incremental energy cost function for CCGT j at configuration y
$i_{j}^{0,x}$	Initial commitment status of PT x for CCGT j
$\underline{P}_{j,t}^{y}/\bar{P}_{j,t}^{y}$	Generation resource minimum/maximum output limit of CCGT j at configuration y and time t
RU_{j}^{y}/RD_{j}^{y}	Ramp up/down rate for configuration y of CCGT j at non-shutdown intervals
$RU0_{j}^{y}$	Ramp up rate for configuration y of CCGT j at startup intervals
$T_{w}^{SU,x}$	Minimum time that PT x must be down for startup type w
TD_{j}^{x}/TU_{j}^{x}	Minimum down/up time of PT x for CCGT j
$TD_{j}^{0,x}/TU_{j}^{0,x}$	Number of time intervals that PT x of CCGT j has been off/on before the scheduling horizon

I. Introduction

A. Motivation

As THE electricity grid accounts for 35% of U.S. total CO_2 emissions in 2016 as the largest contributor, more stringent environmental regulations on power plants become the most critical concern. Indeed, according to the Clean Power Plan 2015, the U.S. power sector targets to reduce 32% from its 2005 CO_2 emission level by 2030 [1], [2]. To achieve this goal, one effective building block is to increase power generation from lower-emission CCGTs.

Compared to traditional thermal units, CCGTs enjoy many advantages, including shorter installation time, lower levelized cost of electricity, faster response time, less CO_2 emissions, as well as higher operation efficiency and reliability [3]–[5]. Indeed, different from coal-fired and nuclear units that are incapable of providing sufficient flexibility, faster ramping capability of CCGTs makes them ideal to support a deeper penetration of highly variable renewable resources such as wind and solar energy. In addition, because of the new horizontal drilling and hydraulic fracturing techniques, huge shale gas reserves have been discovered in the U.S. and natural gas becomes commercially sustainable with a price comparable to that of coal. Therefore, CCGTs have made up majority of the new generation capacities in the last decade, and an even higher share of CCGTs is expected in the near future [6]. For instance, in ISO-NE, roughly 3,000 MW of new CCGTs is coming forward in recent auctions and will be available in 2017-2019 [7].

However, due to complicated operation characteristics, CCGTs bring significant challenges to the market modeling and clearing processes. In fact, for a CCGT, exhausted gas from CTs can be used by HRSG to produce steam and drive STs, while on/off switches of CTs and STs have to follow a required sequence. A CCGT can work on different configurations (modes) with respect to various on/off status combinations of CTs and STs, presenting different physical and economic features, e.g., ramp rates, operation cost, etc. Fig. 1 shows the valid configurations and feasible transitions for a typical CCGT with 2 CTs and 1 ST (denoted as "2 × 1").

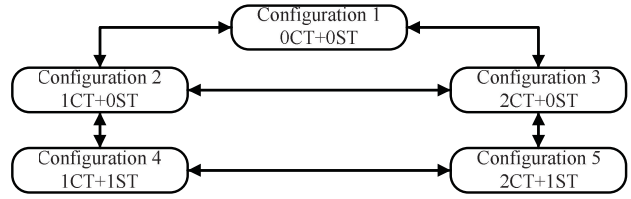

Fig.1. Feasible transition diagram of a CCGT with 2 CTs and 1 ST.

Managing one of the largest electricity markets in the world, MISO encountered computational challenges in solving day-ahead SCUC in early 2014 due to the significant expansion of its market footprint. Specifically, according to the NERC requirement and the stringent market clearing time limitation, it is not practically feasible to model full network constraints for both the base case and post-contingency scenarios. Consequently, only a limited proportion of critical pre- and post-contingency transmission constraints are incorporated in SCUC [8]. In 2017, the MISO market includes 51 CCGTs with the total capacity around 29 GW. MISO received requests from MPs to improve CCGT modelling (e.g., from aggregate model to configuration-based model) in 2011. However, computational challenges prevented MISO from implementing advanced models. Since then, notable research and development (R&D) efforts have been placed to improve computational performance of production SCUC as well as the prototype configuration-based CCGT modeling [8]–[10]. The dedicated R&D effort finally resulted in acceptable computational performance. Meanwhile, MISO continued the research with academia on addressing issues of the configuration-based CCGT model and developed enhanced approaches as proposed in this paper. In October 2017, MISO formed enhanced combined cycle task team and organized monthly meetings to discuss enhanced CCGT modeling with stakeholders. As is agreed in those meetings that operational features of CCGT are described by properties of both configurations and components, modeling options originated from this paper received the majority support. It will be used as the base model for design and implementation [11]. Indeed, better CCGT modelling in market clearing engine could reduce production costs and incentivize MPs to offer true costs.

B. Literature Review

As summarized in Table I, five CCGT models currently exist in academic research and industry practice.

1) Aggregate Model: Aggregate model simplifies each CCGT as a regular unit by allowing the CCGT owner to provide an aggregate offer for the entire CCGT. This model sacrifices the accuracy because the aggregate offer cannot rigorously represent operation costs or physical operating characteristics

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TABLE I
SUMMARY OF CCGT MODELS IN LITERATURE

Model	1	2	3	4	5
Academic research	N/A	[12]	[5], [15]	[4], [5], [10], [13], [15]-[24]	[25],[26]
Industry application	MISO, PJM, NYISO	ISO-NE	N/A	ERCOT, SPP, CAISO, MISO prototype	N/A

of CCGTs [12]. As a result, the SCUC engine may dispatch CCGTs at generation levels that PTs cannot reach [13].

2) Pseudo Unit Model: Pseudo unit model represents a CCGT via multiple pseudo units, each comprising a single CT and an associated portion of the ST capacity. To ensure consistent commitment status of a same ST in multiple pseudo units, all pseudo units are modelled similarly, including startup cost, incremental energy cost, minimum/maximum capacities, and minimum on/off times. Consequently, the pseudo unit model fails to precisely represent these physical characteristics, which are usually different under distinct configurations.

3) Component-Based Model (CPBM): CPBM simulates individual components (i.e., PTs) of CCGTs separately, in order to preserve generation characteristics of PTs and accurately reflect certain important features such as minimum up/down time and startup/shutdown cost. In addition, each PT has a market-clearing dispatch value that can be used for power flow assessment and contingency analysis [15]. However, this model has not been applied in industry practice due to the following issues: (i) the unfriendly offer format for MPs [5], and (ii) the difficulty in modeling certain characteristics of configurations within a CCGT, such as feasible transitions, energy costs, and ramp rates, due to the strong dependency between CTs and STs of a CCGT.

4) Configuration-Based Model (CFBM): In CFBM, different combinations of CTs/STs are considered as mutually exclusive configurations, as shown in Fig. 1. As the most extensively investigated model, CFBM can better reflect operational dependency between CTs and STs of a CCGT, especially incremental energy costs, ramp rates, and feasible transitions between configurations. Additionally, in order to mitigate computational challenges in the market clearing process, practically some ISOs/RTOs limit the number of registered configurations per CCGT [17]. MISO has been evaluating CFBM for several years [9], [10], [16], and benefit study shows that it could potentially bring about $14 to $34 million annual cost saving over the aggregate model [11]. With the dedicated research on computational models and algorithms, MISO has greatly improved performance of the market clearing software with CFBM modeling [10] and starts to consider the implementation [9].

However, treating each configuration as a regular unit while not distinguishing individual PTs underneath each configuration, CFBM may undermine the accuracy of certain PT operation characteristics, such as minimum up/down times and time-dependent transition costs, which could result in inefficiency/infeasibility of the market clearing engine. To better capture minimum up time limits of individual PTs, SPP introduced "group run time constraints" [18], in which each group represents a set of configurations in a CCGT that share the same operation feature. In addition, transition cost is usually related with startup costs of individual PTs [10], and startup warmth type of each configuration is approximated via CCGT "offline" status [19]. All these features are hard to be accurately modeled in CFBM.

5) Edge-Based Model (EBM): Noticing the issues from previous models, EBM [25], [26] was recently proposed to capture physical features of individual PTs such as minimum up/down constraints. It represents a more accurate model than previous ones and also ignites our research. However, different from the logic variables and constraints that was proposed more than half century ago [27] and has been widely deployed in day-ahead market clearing software to describe the relationship between startup/shutdown indicators and commitment statuses, EBM requires an entirely new logic to represent CCGT operations. This could increase modelling complexity and bring extra efforts for implementing the EBM formulation.

C. Contribution and Paper Organization

In order to represent operational features of CCGTs more accurately while alleviating computational burden in day-ahead market, we extend the model in our existing work [28] to propose a novel hybrid CCGT model in this paper. Main contributions include the following aspects.

1) Driven by the needs from MPs, a hybrid CCGT model is proposed by taking advantages of offers from both configurations and individual PTs. The proposed hybrid model can reflect CCGT physical features more accurately, enhance operational flexibility, and avoid operational infeasibility of PTs.
2) Tight formulations are further conducted to relieve computational burden of the proposed hybrid CCGT model. The hybrid model is compared with the extensively studied CFBM in terms of the problem size and potential computational impacts.
3) By making use of commercial MIP solvers, the hybrid model is compared with other CCGT models in terms of computational time. The proposed tight hybrid model shows reasonable computational performance along with great cost saving compared with the tightened CFBM formulation [10], [24]. Compared with the recently proposed EBM [25], [26], the hybrid model also shows better computational efficiency.

The remaining paper is organized as follows. Section II presents the proposed hybrid CCGT formulation. Section III discusses efforts for achieving tighter formulations and better computational performance. Case studies are shown in Section IV, and conclusion is drawn in Section V.

II. MATHEMATICAL FORMULATION OF THE HYBRID MODEL

With features from both configurations and individual PTs, different constraints are respectively formulated according to configurations or individual PTs in the hybrid CCGT model. Indeed, it has been confirmed by MPs that hybrid offers could better describe CCGT operational features. For the sake of concise representation, subscript j for CCGTs is omitted in this section.

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In addition, the full SCUC formulation, especially the transmission network constraints for normal conditions and contingency, is not presented, which can be referred in authors' previous work [8], [29].

A. Mapping Between Configurations and Individual PTs

The MISO system currently includes CCGTs from different manufacturers with significant variations in their components. Thus, for a CCGT with different CTs and/or STs, a complete configuration model is required to depict its behavior. While in practice, most of CCGTs are equipped with resembled CTs/STs. Consequently, configurations with same numbers of CTs and STs could be aggregated to reduce computational burden and render MPs operational flexibility. To introduce the hybrid model, the mapping relationship between configurations and PTs of a CCGT is first revealed, for both complete and aggregate configuration models.

1) Complete Configuration Model: We illustrate the mapping relationship between configurations and individual PTs of a CCGT in the proposed hybrid model by a "2 × 1" example shown in Fig. 1. When the 2 CTs are not identical, i.e., with different incremental energy costs, ramp rates, minimum up/down times, or capacities, 7 configurations are required to represent all valid operating modes. These configurations are all off, $\mathrm{CT}_1, \mathrm{CT}_2, \mathrm{CT}_1+\mathrm{CT}_2, \mathrm{CT}_1+\mathrm{ST}, \mathrm{CT}_2+\mathrm{ST}$, and $\mathrm{CT}_1+\mathrm{CT}_2+\mathrm{ST}$, and their commitment statuses at each time interval, denoted as u_t^1 to u_t^7, are mutually exclusive. In addition, 10 upward transitions between valid configurations in successive hours, with startup switch of certain PT, are feasible, which are denoted as $v_t^{12}, v_t^{13}, v_t^{14}, v_t^{24}, v_t^{25}, v_t^{34}, v_t^{36}, v_t^{47}, v_t^{57}$, and v_t^{67}. Similarly, 10 feasible downward transitions with shutdown switch of certain PT are represented by the reversed superscripts. Finally, the mapping relationship between commitment statuses of PTs i_t^x and configurations u_t^y is expressed as in (1) and denoted as $\boldsymbol{M}$.

$$\boldsymbol{I} = \begin{bmatrix} i_t^{c1} \\ i_t^{c2} \\ i_t^{s1} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_t^1 \\ \vdots \\ u_t^7 \end{bmatrix} = \boldsymbol{M} \cdot \boldsymbol{U} \tag{1}$$

Similarly, (2)/(3) describes the relationship between PTs' startup/shutdown indicators and upward/downward transition variables, where $\boldsymbol{M^{up}}/\boldsymbol{M^{dn}}$ represents mapping relationship between start up/shutdown of PTs and upward/downward transitions between configurations.

$$\boldsymbol{SU} = \begin{bmatrix} su_t^{c1} \\ su_t^{c2} \\ su_t^{s1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_t^{12} \\ \vdots \\ v_t^{67} \end{bmatrix} = \boldsymbol{M^{up}} \cdot \boldsymbol{V^{up}} \tag{2}$$

$$\boldsymbol{SD} = \begin{bmatrix} sd_t^{c1} \\ sd_t^{c2} \\ sd_t^{s1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_t^{21} \\ \vdots \\ v_t^{76} \end{bmatrix} = \boldsymbol{M^{dn}} \cdot \boldsymbol{V^{dn}} \tag{3}$$

Note that $\boldsymbol{M^{up}}$ and $\boldsymbol{M^{dn}}$ could be derived from $\boldsymbol{M}$ if we view constraints row-wise and further consider the relationship between configurations and their feasible transitions. That is to say, only mapping matrix $\boldsymbol{M}$ is needed as input to build constraints (1)–(3). In addition, as only one configuration status u_t^y is chosen at each time interval, a unique transition $v_t^{yy'}$ could be accordingly determined. Thus, after mapping relationships $\boldsymbol{M}$, $\boldsymbol{M^{up}}$, and $\boldsymbol{M^{dn}}$ are constructed, PT-related variables i_t^x, su_t^x, and sd_t^x could be fully represented by configuration-related variables as in (4), where sets ON^x, UT^x, and DT^x can be derived from corresponding rows of $\boldsymbol{M}$, $\boldsymbol{M^{up}}$, and $\boldsymbol{M^{dn}}$.

$$i_t^x = \sum_{y \in ON^x} u_t^y \ ; \ su_t^x = \sum_{yy' \in UT^x} v_t^{yy'} \ ; \ sd_t^x = \sum_{yy' \in DT^x} v_t^{yy'} \tag{4}$$

2) Aggregate Configuration Model: In an aggregate configuration model, for the "2 × 1" example, configurations CT_1 and CT_2 as well as $\mathrm{CT}_1+\mathrm{ST}$ and $\mathrm{CT}_2+\mathrm{ST}$ will be combined. Consequently, the valid configurations are all off, 1CT, 2CT, 1CT+1ST and 2CT+1ST, denoted as u_t^1 to u_t^5, together with 12 valid transitions (6 upward and 6 downward). By further considering that CT_1 has a higher priority than CT_2 when a CT is needed to switch on, constraint (5) describes the mapping relationship between commitment statuses of PTs and configurations. Similarly, constraints (6), (7) describe the mappings between startup/shutdown and feasible transitions, with $v_t^{12}, v_t^{13}, v_t^{23}, v_t^{24}, v_t^{35}, v_t^{45}$ and $v_t^{21}, v_t^{31}, v_t^{32}, v_t^{42}, v_t^{53}, v_t^{54}$ representing upward and downward transitions.

Although priorities of alternative CTs/STs are enforced in the aggregate configuration model, plant operators do have flexibility to switch from one CT/ST to another as long as it meets the scheduled capacity and other physical limits (especially the features related to warmth status and minimum up/down time constraints) are not violated. This is valuable when one of alternative PTs encounters physical failures.

$$\begin{bmatrix} i_t^{c1} \\ i_t^{c2} \\ i_t^{s1} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_t^1 \\ \vdots \\ u_t^5 \end{bmatrix} \tag{5}$$

$$\begin{bmatrix} su_t^{c1} \\ su_t^{c2} \\ su_t^{s1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_t^{12} \\ \vdots \\ v_t^{45} \end{bmatrix} \tag{6}$$

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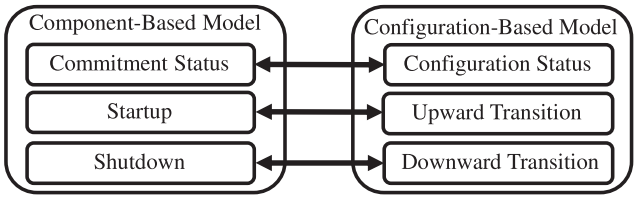

Fig. 2. Relationship between CPBM and CFBM.

$$\begin{bmatrix} sd_t^{c1} \\ sd_t^{c2} \\ sd_t^{s1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_t^{21} \\ \vdots \\ v_t^{54} \end{bmatrix} \tag{7}$$

3) Remarks: As discussed above and further illustrated in Fig. 2, there are two sets of elements in CCGT models, i.e., PTs and configurations. Indeed, CPBM focuses on the left subgraph of Fig. 2 with only PT-related commitment statuses, startup, and shutdown indicators, while it is hard to only use PT-related variables to describe transition logics between configurations. In comparison, CFBM focuses on the right subgraph of Fig. 2 with configuration-related commitment statuses and transition indicators, in which operational characteristics of individual PTs are largely neglected.

In fact, these two sets of elements can be distinctly mapped to each other. That is, variables in CFBM contain sufficient information to retrieve commitment statuses of individual PTs. Reversely, u_t^y and $v_t^{yy'}$ can also be linearly expressed via PT variables. However, unlike configuration-related variables, PT-related variables are not exclusive at each time interval, and in turn binary nature of configuration-related variables may not hold via a simple linear mapping from PT-related variables. That is, u_t^y and $v_t^{yy'}$ could be integers instead of binary variables, which deviate from their physical meanings. Actually, the range of integer values corresponding to a PT (when it is on) defines the feasible transition information required in CPBM (i.e., constraints (20)–(30) and input information of Table IV in [5]).

In summary, with the established mapping relationships, variables in CFBM are sufficient to describe statuses of PTs. Therefore, the proposed hybrid model remains a tight formulation with respect to the search space of variables as those of tight CFBM [10], [24].

4) Duct Burner (DB) Modelling: DB is a common technology deployed by CCGT owners to enhance energy utilization and increase power outputs. DB uses supplementary firing to increase heat energy of a gas turbine's exhaust, in an attempt to increase power output of a downstream HRSG and result in an extended ST capacity. Data from MPs shows that 14 out of 31 (45.2%) CCGTs in MISO are equipped with DBs. In addition, we also notice some unusual spikes in last segment of certain configuration energy offers, which imply that some MPs may submit combined offers with DBs and regular configurations. Thus, the actual number of CCGTs in MISO market that are equipped with DBs might be even higher.

Indeed, DB can be modeled as an extra PT in the hybrid model. Other power enhancement approaches of CCGTs discussed in [3], [5], [12] could also be handled similarly, i.e., by adding associated configurations and virtual PTs, followed by the predefined transition matrix and mapping relationship. Table II displays a typical transition matrix of a CCGT with DB in MISO market.

TABLE II
TYPICAL TRANSITION MATRIX OF A CCGT WITH DB

	1CT1ST	1CT1ST1DB	2CT1ST	2CT1ST1DB	All Off
1CT1ST	1	1	1	0	1
1CT1ST1DB	1	1	0	0	0
2CT1ST	1	0	1	1	1
2CT1ST1DB	0	0	1	1	0
All Off	1	0	1	0	1

B. Constraints With Configuration Features

A CCGT could operate at different configurations as a pseudo thermal unit. Similar to CFBM, the following operation cost and physical constraints are considered closely related to the configuration features.

1) Incremental Energy Cost: Operation cost of each CCGT $C^{P,y}(p_t^y)$ is modeled via configuration status as shown in (8). Configuration-based offers could be more accurate to reflect operation features of CTs/STs. That is, as output of an ST is strongly related to that of a CT, energy cost of the entire CCGT is not a simple summation of CTs and STs while ignoring their dependency. Here, locally ideal formulation with special ordered set of type 2 (SOS2) is used to piecewise linearize the energy cost function $C^{P,y}(p_t^y)$ [32], which has been proved to be the convex envelope of piecewise linear cost function, and test results on the MISO system have shown great computational benefits [10].

$$\sum_{t\in T}\sum_{y\in Y}\left[C^{P,y}\left(p_t^y\right)\right] \tag{8}$$

2) Exclusion of Configurations: For each time t, all valid configurations of a CCGT are exclusive, as shown in (9).

$$\sum_{y\in Y} u_t^y = 1 \qquad \forall t \tag{9}$$

3) Feasible Transition Between Configurations: Similar to startup and shutdown indicators of regular units [27], $\sum_{y'\in F^{*,\,y}} v_t^{y'y} / \sum_{y'\in F^{y,\,*}} v_t^{yy'}$, which refers to summation of all feasible transitions to/from configuration y at t, can be regarded as startup/ shutdown transition of configuration y at t [10], [20], [24]. Thus, the logic relationship of commitment status for configuration $\boldsymbol{y}$ between two consecutive time intervals can be expressed as in (10).

$$u_t^y - u_{t-1}^y = \sum_{y'\in F^{*,y}} v_t^{y'y} - \sum_{y'\in F^{y,*}} v_t^{yy'} \quad \forall t, y\in Y \tag{10}$$

4) Capacity Limit: Each configuration of a CCGT, similar to regular units, has minimum/maximum capacity limit $\underline{P}_t^y / \bar{P}_t^y$ as shown in (11).

$$u_t^y \cdot \underline{P}_t^y \le p_t^y \le u_t^y \cdot \bar{P}_t^y \qquad \forall t, y\in Y \tag{11}$$

5) Ramping Constraints: As different combinations of CTs and STs may have distinct ramp up/down rates, ramp rate is

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considered as a feature more closely associated with configuration status as shown in (12), (13) [10]. In (12), if there is no transition to configuration $\boldsymbol{y}$, i.e., $\sum_{y' \in F^{*,y}} v_t^{y'y} = 0$, ramp up constraint is ruled by $p_t^y - p_{t-1}^y \leq RU^y$; if a CCGT transits to configuration y, i.e., $\sum_{y' \in F^{*,y}} v_t^{y'y} = 1$, (12) becomes $p_t^y - p_{t-1}^y \leq \sum_{y' \in F^{*,y}} v_t^{y'y} RU0^y$. Same implications apply to ramp down constraint (13). Indeed, ramp constraints (12), (13) present better computational performance in the MISO system with asymmetric startup/shutdown features. To be more specific, shutdown cost is not counted and shutdown ramping is not required in the MISO system. For instance, (13) implies that a resource can move from any $p_{t-1}^y \leq \bar{P}_{t-1}^y$ to $p_t^y = 0$ in a shutdown period t. It is noteworthy that constraints (12), (13) could be replaced by other tight forms such as those in [24] for CCGTs with general ramping requirements.

$$p_t^y - p_{t-1}^y \leq RU^y + \sum_{y' \in F^{*,y}} v_t^{y'y} \left(RU0^y - RU^y\right) \qquad \forall t, y \in Y \quad (12)$$

$$p_{t-1}^y - p_t^y \leq u_t^y RD^y + \left(1 - u_t^y\right) \bar{P}_{t-1}^y \qquad \forall t, y \in Y \quad (13)$$

C. Constraints With Individual PT Features

Working at different configurations though, CCGTs still retain certain features of individual PTs that can be modeled via following physical constraints and costs.

1) Minimum Up/Down Time: Minimum up/down time constraints of individual PTs are expressed as in (14), (15). Same as regular units, [33] showed that (14), (15) represent facet projections of min-up/down constraints in the space of (i^x, su^x, sd^x). Replacing PT-related variables by (4), constraints (14), (15) are equivalently expressed as (16), (17).

$$\sum_{t'=t-TU^x+1}^{t} su_{t'}^x \leq i_t^x \qquad \forall t \in [TU^x, T], x \in X \quad (14)$$

$$\sum_{t'=t-TD^x+1}^{t} sd_{t'}^x \leq 1 - i_t^x \qquad \forall t \in [TD^x, T], x \in X \quad (15)$$

$$\sum_{t'=t-TU^x+1}^{t} \sum_{yy' \in UT^x} v_{t'}^{yy'} \leq \sum_{y \in ON^x} u_t^y \qquad \forall t \in [TU^x, T], x \in X \quad (16)$$

$$\sum_{t'=t-TD^x+1}^{t} \sum_{yy' \in DT^x} v_{t'}^{yy'} \leq 1 - \sum_{y \in ON^x} u_t^y \qquad \forall t \in [TD^x, T], x \in X \quad (17)$$

As constraints (16), (17) describe min-up/down time of each PT independently, they could be added separately to alleviate computational burden if a certain PT is dominant in a CCGT or multiple CTs/STs within the same CCGT present same min-up/down time requirements. This strategy is applied in the MISO system study in Section IV.B.

In addition, similar to regular units, the required initial online time $TU^{R,x}$ and offline time $TD^{R,x}$ of each PT due to the min-up/down time requirements are expressed as in (18), (19). Corresponding configuration commitment variables are fixed in those initial periods (20).

$$TU^{R,x} = \max\left\{0, \left(TU^x - TU^{0,x}\right) \cdot i^{0,x}\right\} \qquad \forall x \in X \quad (18)$$

$$TD^{R,x} = \max\left\{0, \left(TD^x - TD^{0,x}\right) \cdot \left(1 - i^{0,x}\right)\right\} \qquad \forall x \in X \quad (19)$$

$$\sum_{y \in ON^x} u_t^y = i^{0,x} \qquad \forall t \in \left[1, TU^{R,x} + TD^{R,x}\right], x \in X \quad (20)$$

2) Time-Dependent Startup Cost: Time-dependent startup cost is usually applied on regular units [30], [34], and is related to how long the unit has been off. Up to three startup types are modeled in CFBM for the MISO system [10], including hot, intermediate, and cold. As this feature is strongly associated with individual PTs, after discussing with MPs and plant operators, it is agreed that such cost should be more accurately described via PTs' startup cost instead of transition cost between configurations as in CFBM. Constraints (21), (22) are used to determine startup types. Note that $T_1^{x,SU}$ is set to TD^x to describe the hot startup time zone. Similarly, su_t^x and sd_t^x could be substituted via constraint (4), and constraints (21), (22) can be equivalently expressed as in (23), (24). In turn, startup cost of a CCGT is described as $\sum_{x \in X} \sum_{t \in T} \sum_{w \in W} C_w^{SU,x} \cdot \delta_{t,w}^x$.

$$\delta_{t,w}^x \leq \sum_{t'=T_w^{x,SU}}^{T_{w+1}^{x,SU}-1} sd_{t-t'}^x, \quad \forall t \in \left[T_{w+1}^{x,SU}, T\right], w = \{1,2\}, x \in X \quad (21)$$

$$\sum_{w \in W} \delta_{t,w}^x = su_t^x \qquad \forall\, t, x \in X \quad (22)$$

$$\delta_{t,w}^x \leq \sum_{t'=T_w^{x,SU}}^{T_{w+1}^{x,SU}-1} \sum_{yy' \in DT^x} v_{t-t'}^{yy'} \qquad \forall t \in \left[T_{w+1}^{x,SU}, T\right], w = \{1,2\}, x \in X \quad (23)$$

$$\sum_{w \in W} \delta_{t,w}^x = \sum_{yy' \in UT^x} v_t^{yy'} \qquad \forall\, t, x \in X \quad (24)$$

For a PT x having been initially off for $TD^{0,x}$ hours, startup type during hours $(T_{w+1}^{x,SU} - TD^{0,x}, T_{w+1}^{x,SU})$ are fixed to zero as in (25).

$$\delta_{t,w}^x = 0, \forall t \in \left(\max\left\{0, T_{w+1}^{x,SU} - TD^{0,x}\right\}, T_{w+1}^{x,SU}\right), \quad w = \{1,2\}, x \in X \quad (25)$$

Furthermore, for transitions involving startup of multiple PTs, e.g., from "all off" to 2 CTs, startup costs of individual PTs can be calculated separately since their on/off time durations might be different. In fact, for the "2 × 1" example in Fig. 1, up to 9 different transition costs can be modeled with respect to 3

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types of CT startup costs, which provides additional modelling capability than CFBM [10].

3) Additional Specific Constraints: With the capability of modeling properties of individual PTs, certain specific constraints in the MISO system, such as maximum daily start time and energy limits that have been enforced on configurations in CFBM [10], can also be applied on individual PTs to meet practical operation requirements.

D. Computational Complexity Comparison With CFBM

This section analyze scale of the proposed hybrid model and related computational complexity by comparing with CFBM. The proposed hybrid model is formulated via constraints (8)–(13), (16)–(20) and (23)–(25). Compared with CFBM in [10] and [24], constraints (16)–(20) and (23)–(25) represent the major difference between these two models. However, considering the fact that the hybrid model and CFBM are built on different input data, i.e., distinct min-up/down time and time-dependent startup cost for PTs and configurations respectively, it would be difficult to fairly and accurately quantify their integrated impacts on formulation tightness. Alternatively, the comparison is conducted with respect to following key parameters that are well understood as predictors for computational complexity of MIP problems, although in certain instances these parameters may be inadequate [35]. Indeed, computational performance of the two models could also be strongly related with specific input data, and this issue will be further illustrated via numerical case studies in Section IV.A.2.

- *Number of Binary Variables*: With the mapping relationship (4) between configurations and PTs erected, no additional binary variables are required for representing min-up/down behavior of individual PTs. As for time-dependent startup indicators, because the number of PTs is usually less than that of upward transitions, the hybrid model would encounter fewer binary variables and in turn may result in less computational complexity. For instance, the aggregated "2 × 1" example includes 3 PTs v.s. 6 upward transitions, which results in 15 binary variables (9 startup indicators and 6 upward transition variables) in the hybrid model v.s. 18 transition variables (6 upward transitions × 3 startup types) in CFBM [10].
- *Number of Constraints*: Numbers of constraints for min-up/down time and time-dependent startup cost are proportional to the number of PTs in the hybrid model, whereas proportional to the number of configurations in CFBM. As the number of PTs is usually smaller than that of configurations, e.g., 3 PTs v.s. 5 configurations in the aggregated "2 × 1" example, this indicates a problem with fewer constraints.
- *Number of Non-zeros:* As behavior of individual PTs is indicated by the summation of configuration-based variables, more non-zeros are introduced per constraint, which could impact tightness of the formulation [35], [36]. Specifically, compared with min-up/down time constraints of CFBM, constraints (16), (17) for each PT introduce more non-zeros, while the total numbers of non-zeros in the two models would also depend on numbers of PTs and configurations per CCGT as well as their min-down/up time requirements.

III. Choice of Tight Formulations

This section analyzes different formulation choices in the hybrid model for achieving better computational performance. Inspired by state-of-the-art research on formulation techniques of general unit commitment (UC) and CCGT modeling [24], [30]–[34], [37]–[39], as well as our extensive computational studies on the MISO system, the following issues of the proposed hybrid CCGT model are discussed.

A. Choice of Feasible Transition Constraints

To formulate feasible transitions of configurations, it is straightforward to deploy the logic similar to constraint (10) in CFBM for describing relationship between startup/shutdown indicators and commitment statuses of individual configurations. Alternatively, with the capability to model behaviors of individual PTs, feasible transition could also be enforced with PT startup/shutdown logics as in (26), and further expressed as in (27) by leveraging constraint (4).

$$i_t^x - i_{t-1}^x = su_t^x - sd_t^x \qquad \forall t, x \in X \tag{26}$$

$$\sum_{y \in ON^x} u_t^y - \sum_{y \in ON^x} u_{t-1}^y = \sum_{yy' \in UT^x} v_t^{yy'} - \sum_{yy' \in DT^x} v_t^{yy'} \qquad \forall t, x \in X \tag{27}$$

Indeed, respectively derived from logics of configurations and PTs, constraints (10) and (27) could be deployed exclusively or concurrently to obtain valid MIP models that are mathematically equivalent but potentially with different computational performance. Specifically, compared with (10), constraint (27) involves fewer constraints as the number of PTs is usually less than that of configurations, while more non-zeros could be induced in each constraint by comparing $\sum_{y \in ON^x} u_t^y$ to a single u_t^y in (10). In terms of the total number of non-zeros per CCGT from constraints (10) and (27), in the aggregated "2 × 1" example, the values are 34 and 30 for (10) and (27), respectively. Indeed, the compactness/tightness may vary according to valid configurations and feasible transitions predefined for a specific CCGT as well as the minimum up/down time data, therefore a universal conclusion on the integrated computational impact may not be readily available. In addition, although constraint (27) is not necessary from the model validity point of view, extensive case studies show that including (27) into the model could potentially help the MIP presolving process [41], [42], with a better root-node relaxation value (better lower bound) that is closer to the optimal integer solution, and also the cutting plane generation process from MIP solvers [35], [40], [42]. Computational performance of (27) and (10) will be compared and discussed via numerical case studies in Section IV.A.3.

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B. Exclusion of Transition Process

For each CCGT, constraint (28) could be formulated to describe that at most one feasible transition is allowed in a CCGT at each time interval. Constraint (28) is known as a clique inequality [35], [36], [41] from set packing constraints, where all feasible transitions of a CCGT at each time interval consists a maximum clique. This type of constraint could tighten the lower bound of proposed model and significantly shorten computational time by preventing the exploration of unnecessary branches.

$$\sum_{y \in Y} \sum_{y' \in F^{y,*}} v_t^{yy'} \leq 1 \qquad \forall\, t \tag{28}$$

Constraint (28) applies to both CFBM and the hybrid model, but presents different impacts on formulation tightness. Specifically, in CFBM, as constraint (28) is dominated by the min-up/down time constraints on the configuration and constraints (9), (10), adopting (28) in CFBM may not further tighten formulation or improve lower bound, but is mainly from practical computational concerns. Indeed, as some units/ configurations do not have min-up and/or -down parameters, adapting constraint (28) in CFBM can tighten the model and prevent unnecessary search branches by reducing the number of vertices as indicated in [10]. However, different from CFBM, as min-up/down time constraints are modeled on PTs in the hybrid model, constraint (28) becomes critical to cut off fractional solutions of LP relaxation problem and tighten the lower bound. The tightness of including constraint (28) in the hybrid model is further illustrated by an example in Appendix.

C. Choice of Binary/Continuous Variables

$v_t^{yy'}$ represents the transition between configurations. As shown in (10), similar to startup/shutdown variables in UC, $v_t^{yy'}$ would only take binary values even defined as continuous variables. However, numerical test between 3-bin and 1-bin variable formulations on UC [37], [38] show that 3-bin could have better computational performance, because additional binary variables would induce better branching choices or the improved heuristic approach in presolving process of the MIP solvers. Computational performance of declaring $v_t^{yy'}$ as binary and continuous variables will be compared in Section IV.A.3.

In addition, declaring time-dependent startup indicators as binary or continuous variables would hold same integrality property as suggested in [30], [34], as startup indicators are strictly limited by shutdown indicator and impacted by unit's offline times. Related computational performance will be compared in Section IV.B.

D. Approach to Reduce the Number of Time-Dependent Startup Indicators

The number of time-dependent startup indicators could be reduced if startup costs of different startup types are the same. Specifically, for PTs with equal intermediate and hot startup costs $\delta_{t,1}^x$ can be removed, while $\delta_{t,2}^x$ can be removed for transitions with equal cold and intermediate PT startup costs. This technique is also applicable to regular units for reducing the number of time-dependent startup indicators.

TABLE III
CONFIGURATION STATUS OF THE CCGT AT BUS 49

Mode	Hours (1-24)
Case 1.1	1 1 2 2 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
Case 1.2	**2 2** 2 **4** 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
Case 2.1/2.2	1 1 **1 1 2** 4 **4** 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5

IV. CASE STUDIES

In this section, the proposed hybrid model is implemented and compared with CFBM on the modified IEEE 118-bus [43] and the MISO system. Comparative analysis on economic impact and computational performance with CFBM is first conducted via the modified IEEE 118-bus system, along with the computational comparison with EBM by using state-of-the-art commercial MIP solvers. Effectiveness of the tight formulations discussed in Section III is further analyzed on the two systems with extensive studies.

A. The Modified IEEE 118-Bus System

The modified IEEE 118-bus system is studied to compare performance of the proposed hybrid model and CFBM in [24] with respect to different min-up/down time limits. CFBM in [24] has been proved to be tight and computationally efficient. Besides, constraint (28) is applicable to operation logic of both models and implemented to tighten both formulations. 8 CCGTs with the aggregated "2 × 1" configuration along with 54 regular thermal units are studied, and their initial statuses are all set to offline for sufficient long period. Cost functions of regular units and CCGTs are piecewise linearized into 5 segments. The following four cases are studied and solved by Gurobi 6.5.2 [44] on a personal laptop with Intel Core i7 2.6 GHz CPU and 12 GB memory.

- *Case 1.1/1.2:* CFBM with configuration transition costs and min-up and down times of 2/3 hours for all configurations.
- *Case 2.1/2.2:* Hybrid model with turbine startup/shutdown costs and min-up and down times of 2/3 hours for all PTs.

1) Economic Comparison With CFBM: Commitment status of the CCGT at Bus 49 shown in Table III is used to illustrate operational flexibility of the hybrid model. Specifically, in order to transit from offline status to the most economic Configuration 5, in Cases 1.1 and 1.2, the CCGT needs to stay in Configurations 2 and 4 for 2 and 3 hours respectively, while all transitions only involve the startup of PTs and do not incur min-up/down violations from individual PTs' perspective. In comparison, both scenarios in Case 2 have the same commitment result with respect to different PT min-up/down time limits, which shows effectiveness of the proposed hybrid model by taking advantage of accurate formulation on physical characters of individual PTs to identify the most cost efficient schedule. Comparison on costs of the two models in Table IV shows savings in production cost under the hybrid model. In addition, for Case 2, as only startup

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TABLE IV
COST COMPARISON ON THE MODIFIED 118-BUS SYSTEM

	Case 1.1	Case 1.2	Case 2.1	Case 2.2
Obj ($)	1,511,836.81	1,514,848.79	1,508,854.85	1,508,854.85

TABLE V
COMPUTATIONAL TIME OF CFBM AND THE HYBRID MODEL (S)

Solver	Gurobi			CPLEX		
Gap	0.5%	0.1%	0.01%	0.5%	0.1%	0.01%
Case1.1	57	68	151	9	25	173
Case1.2	50	65	73	11	33	38
Case2.1	57	65	97	11	32	163
Case2.2	69	75	105	8	32	233

is involved and no PT of any CCGT is shutdown till the end of horizon, both scenarios in Case 2 have the same commitment result and therefore the same total cost.

Indeed, with the capability of modeling min-up/down time limits for individual PTs, the proposed hybrid model could not only achieve lower cost, but also provide more flexibility for both the system and MPs. From the system's point of view, as a CCGT may only involve consecutive upward/downward transitions in daily schedule due to its fast response capability for mitigating hourly load fluctuations, the hybrid model could avoid setting the CCGT in a certain configuration for unnecessary long time to meet the configuration up/down time requirement. It will also allow MPs to offer the true costs and operational parameters.

2) Computational Comparison With CFBM: Results on computational performance of Cases 1 and 2 are listed in Table V with three different optimality gaps, i.e., 0.5%, 0.1%, and 0.01%. Results from CPLEX 12.6.3 [45] are also reported for comparison. As stated in Section II.D, since different offers adopted in the two models may impact their computational performance, while it is hard to equivalently convert information from one to the other, a universal conclusion on computational performance of the two models may not be easily derived. Nevertheless, it is observed in both models that different min-up/down times could greatly impact the selection on binary variables and the presolving procedure in a solver [33], [41], [42], which consequently present heterogeneous computational performance. Through extensive experiments, we observe that different load profiles, cost functions, choice of solvers (type/version), and optimality gap thresholds could all significantly impact the model preference in terms of computational performance.

3) Comparison of Different Hybrid Model Formulations: Different formulations of the hybrid model discussed in Section III are further evaluated. Specifically, F-1 denotes the hybrid model discussed in Section II with constraints (8)–(13), (16)–(20), (23)–(25), (28); F-2 is the model by replacing constraint (10) in F-1 with (27); F-3 refers to the model by adding (27) in F-1; F-4 is the model by excluding constraint (28) from F-1; F-5 includes all constraints of F-1, while transition variables are declared as continuous. Constraints adopted in these five models are listed in Table VI for quick reference.

Computational performances of Cases 2.1-2.2 with different formulations are summarized in Tables VII–VIII, in which "R" indicates computational time for root node relaxation, and "Nodes" denotes number of nodes explored when the predefined optimality gap is reached. LPobj denotes objective value of the LP relaxation, and integer gap (Igap) is defined as $(\mathrm{MIPobj} - \mathrm{LPobj})/\mathrm{MIPobj} \times 100\%$. Meanwhile, Robj denotes objective value of the root-node relaxation (best bound), and root-node relaxation gap (Rgap) is defined as $(\mathrm{MIPobj} - \mathrm{Robj})/\mathrm{MIPobj} \times 100\%$. Root-node relaxation value is derived after the presolving process of an MIP solver. As shown in Table IV, MIPobj in both cases is $1,508,854.85.

TABLE VI
DIFFERENT FORMULATIONS TESTED ON THE 118-BUS SYSTEM

Formulation	Constraints
F-1	(8)-(13), (16)-(20), (23)-(25), (28)
F-2	(8)-(9), (11)-(13), (16)-(20), (23)-(25), (27)-(28)
F-3	(8)-(13), (16)-(20), (23)-(25), (27)-(28)
F-4	(8)-(13), (16)-(20), (23)-(25)
F-5	(8)-(13), (16)-(20), (23)-(25), (28) with $v \in [0,1]$

TABLE VII
COMPUTATIONAL COMPARISON OF DIFFERENT FORMULATIONS

		R(s)	0.5%		0.1%		0.01%	
			Time (s)	Nodes	Time (s)	Nodes	Time (s)	Nodes
Case 2.1	F-1	27	57	27	65	286	97	2979
	F-2	14	45	27	59	638	710	58484
	F-3	28	65	15	72	255	91	2076
	F-4	31	60	27	102	468	2195	154768
	F-5	29	51	28	65	569	1090	51211
Case 2.2	F-1	29	69	28	75	158	105	2911
	F-2	22	53	29	87	739	1396	93230
	F-3	27	63	27	70	248	85	1440
	F-4	25	55	27	159	1415	1430	124831
	F-5	25	47	29	61	628	677	50932

TABLE VIII
COMPUTATIONAL COMPARISON OF DIFFERENT FORMULATIONS (CONT.)

		LPobj ($)	Igap (%)	Robj ($)	Rgap (%)	# Cuts
Case 2.1	F-1	1500698.77	0.5405	1501403.72	0.4938	461
	F-2	1497334.92	0.7635	1500552.28	0.5503	236
	F-3	1500698.77	0.5405	1501497.95	0.4876	429
	F-4	1500680.21	0.5418	1500680.21	0.5418	345
	F-5	1500698.77	0.5405	1500698.77	0.5405	162
Case 2.2	F-1	1500698.77	0.5405	1501316.05	0.4996	512
	F-2	1497334.92	0.7635	1500552.28	0.5503	202
	F-3	1500698.77	0.5405	1501409.69	0.4934	522
	F-4	1500680.21	0.5418	1500680.21	0.5418	194
	F-5	1500698.77	0.5405	1500698.77	0.5405	115

Comparison between F-1 and F-2 shows that constraints (10) and (27), although exploiting same logics to enforce feasible transitions, present varying computational impact. Indeed, it takes less time for F-2 to solve the LP relaxation mainly due to its compactness (fewer constraints and total non-zeros), as indicated by a shorter time to solve the root node relaxation problem and to search for a similar number of nodes, which could result in less total time to reach a relatively larger gap (0.5% and 0.1%). However, for a smaller gap threshold, the effort could be significantly increased as indicated by the time and the number of nodes explored. This could also be illustrated by formulation tightness as indicated in Table VIII, i.e., F-2 derives larger Igap and Rgap that indicate a less tight formulation.

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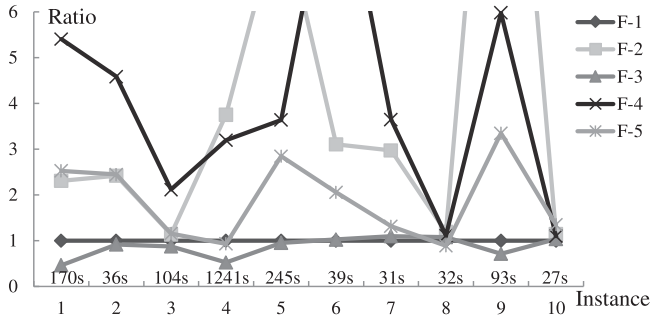

Fig. 3. Comparison of F-1 to F-5 via multiple instances.

Results of F-3 further show the efforts by adding both (27) and (10), which could improve performance with less time and fewer explored nodes in most cases as compared to F-1. From Table VIII, although Igap of F-3 is the same as that of F-1 and the LP relaxation is not further tightened with constraint (27), the root-node relaxation is improved with a lower Rgap after presolving process of the MIP solve. That is, by adding (27), F-3 provides a better solution of the root-node relaxation that is closer to the optimal integer solution. It could always be difficult to identify the root cause of such variabilities in the solution process [46], as presolving process involves plenty of techniques [41]. However, it is reasonable to speculate that these equations would help with non-zero cancellation and/or parallel row detection in redundancy detection of the presolving process [41], [42].

F-4 shows that, when constraint (28) is excluded, computational effort increases tremendously to reach a smaller gap. This could also be explained from both Igap and Rgap that excluding (28) could sacrifice tightness of the LP relaxation. Indeed, it performs a very similar trend as F-2.

Setting transition variables as continuous in F-5 results a formulation with fewer binary variables. This could potentially save time for computing LP relaxations as indicated by more nodes being explored in less time with a larger optimality gap, but at a cost of cumbersome effort in searching for solutions of higher quality with smaller gaps. Indeed, declaring transition variables as integer could enhance performance of cutting plane generation and branching choice of the MIP solver. To be specific, in Case 2.1, numbers of generated cutting planes from Gurobi for F-1 and F-5 are 461 and 162, while are respectively 512 and 162 in Case 2.2. Indeed, numbers of different types of cutting planes differ significantly, especially implied bounds, cover, clique, and flow cover cuts [35], [42]. In addition, comparison between F-1 and F-5 in Table VIII further shows that declaring transition variables as integer also helps the presolving process of the MIP solver, i.e., with a better objective value of relaxed root-node and a smaller Rgap.

To further compare the impact of different formulations F-1 to F-5, ten instances with different load profiles are tested. The relative MIP gap is set to 0.01% as default. Fig. 3 shows computational time in ratio of different formulations as compared to F-1. Computational time of F-1 for different instances are listed above the x-axis of Fig. 3, ranging from around 30 seconds to more than 1000 seconds. Overall, the performance is consistent with the above analysis. Specifically, formulation F-3 with both constraints (27) and (28) outperforms F-1 in most instances, and the benefit is more significant for the ones with longer computational time (i.e., instances 1, 4, and 9).

TABLE IX
COMPUTATIONAL RESULTS OF EBM AND THE HYBRID MODEL IN CASE 2.1

	EBM	Hybrid Model
# of Constraints	40092/17814	42332/19637
# of Variables	23136/20685	22944/20578
# of Non-zeros	855470/241576	863016/226374
# of Binary variables	7152/6341	7152/6242
R (Time/Iterations)	32s/59704	28s/55595
0.5% (Time/Nodes)	73s/29	65s/15
0.1% (Time/Nodes)	85s/611	72s/255
0.01% (Time/Nodes)	232s/10001	91s/2076
MIPObj ($)	1508854.85	1508854.85
Igap (%)	0.4654	0.5405
Rgap (%)	0.4654	0.4876
# of Cutting Planes	70	566

TABLE X
COMPUTATIONAL SUMMARY OF TEN INSTANCES FOR EBM AND THE HYBRID MODEL

	Average Ratio (H/E)
Relaxation Time	0.7570
0.5% Gap Time	0.7215
0.1% Gap Time	0.9022
0.01% Gap Time	0.9020
Igap	1.3198
Rgap	1.0906
# of Cutting Planes	4.0273

In summary, a less tight formulation like F-2 may save effort for deriving an integer feasible solution, but could bring high risks of failing to clear the market within the desired optimality gap and time limit due to excessive branch searching. This could all bring severe issues in practice. Therefore, it is better to deploy constraint (10) instead of merely (27), and set transition variables as binary for tighter formulation. In addition, as constraint (27) and (28) both help tighten the formulation and/or improve root node relaxation, they are also suggested and formulated in the MISO system for achieving better computational performance.

4) Computational Comparison With EBM: In this subsection, the tightened hybrid formulation F-3 is compared with the strengthened EBM [25], [26]. The strengthened EBM is modeled by replacing min-up/down time constraints in [25] with constraints (24) and (39) in [26], and replacing ramping constraints in [25] with constraints (40), (41) in [26]. In addition, different from [26], extra inequalities (i.e., constraints (42)–(50) in [26]) are not implemented in this comparison study because these cuts could significantly increase the problem size and modeling complexity. These additional cuts should be dynamically introduced into the formulation during the solving process, e.g., by "callback" procedure [44], [45].

Table IX lists computational statistics of the two models for Case 2.1, where numbers of constraints, variables, nonzeros, and binary variables include values before/after the presolving process of solvers. Table X calculates average ratios of various factors in the two models for ten instances studied in

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Section IV.A.3, where H/E denotes ratio of the hybrid model over EBM. As both models are able to exploit CCGT features from configurations and PTs, they result in same economic efficiency with the same scheduling of CCGTs and regular units.

The major difference between the two models lies in that: they deploy different logics to depict feasible transitions of CCGT. That is, although the same number of binary variables is used in both models (see # of binary variables before presolving), constraints for depicting feasible transitions of CCGT present different physical meanings. To be specific, different from the hybrid model, the concept of incoming, outgoing, and self-loop edges is used in EBM to represent state transition process among different configurations, while commitment statuses of configurations are implicitly represented by summation of variables corresponding to incoming and self-loop edges. This significantly increases the number of non-zeros in constraints that contain configuration commitment status variables, e.g., capacity limit constraints. Moreover, binary variables of different physical meanings adopt different logics constraints to represent feasible transitions of CCGTs (see logic constraints (1), (2) in [25] and constraints (9), (10), (27), (28) in the proposed hybrid model), and consequently result in formulations before/after presolving of different sizes, especially in terms of the numbers of binary variables and non-zeros.

In general, the proposed hybrid model outperforms EBM in terms of computational times for solving the root-node relaxation and calculating final solutions with respect to different optimality gap thresholds. It is also interesting to note that, although both Igap and Rgap of EBM are smaller than the proposed hybrid model, the overall computational performance is worse. Indeed, a formulation with a lower integrality gap could lead to a faster solution time in theory, whereas computational performance in practice is also highly influenced by cutting planes, heuristics, and presolving techniques adopted in the MIP solver [41], [42], [47]. A significant difference of the two models is that the number of cutting planes generated from the hybrid model is much larger than that from EBM in all ten instances, with an average ratio of 4.0373 for the ten instances. That being said, although suffering a slightly higher integer gap, the logic constraints adopted in the proposed hybrid model are more advantageous to the solver in generating more effective cutting planes and further tightening feasible regions of the original formulation after the presolving process. In addition, root-node relaxation calculation time of the hybrid model is notably shorter than that of EBM, as fewer non-zeros for representing commitment status of configurations could save computational effort for solving the LP relaxation problem (i.e., with fewer simplex iterations). Therefore, the hybrid model is more computationally efficient than EBM with the help of built-in techniques in state-of-the-art commercial MIP solvers.

Moreover, different from the conventional CCGT feasible transition formulation [27] that has been widely deployed in day-ahead market clearing software to describe the relationship between startup/shutdown indicators and commitment statuses, the logic of describing feasible transitions in EBM presents additional implementation complexity, especially setting up the edge-related logic sets (see nomenclature in [25]). In addition, as edges essentially represent state transition processes, this edge-based concept may distract practitioners from the physical meaning when constructing the model. These issues are not necessarily related to computational comparison over the two models, but present practical implementation concerns that should not be neglected.

TABLE XI
SUMMARY ON COMPUTATIONAL PERFORMANCE: AVERAGE VALUES OF DIFFERENT HYBRID MODELS OVER CFBM

Model	Cost Saving ($)	R Time (Ratio)	Time (Ratio)
HM-1	8992.9	1.0553	0.9915
HM-2	10735.1	1.0498	0.8907
HM-3	6495.4	0.9538	1.0952
HM-4	11974.8	1.1333	0.8865

B. The MISO Test Case

To facilitate CFBM cost/benefit study [10], MISO has requested MPs to submit configuration-based offers in early 2017 [9]. Data of 31 CCGTs are collected, with a total of 119 configurations consisted by 100 PTs (including DBs). In addition, some CCGTs require a CT accompanied with an ST due to physical structure requirement (i.e., the ST is always combined with a CT in the aggregated configuration), which further reduces the total number of combined turbines to 74. After that, additional PT data are requested for the proposed hybrid model, including min-up/down time and three types of PTs' time-dependent startup costs/times. Ten 36-hour-ahead load profiles S1–S10 are randomly selected for comparing costs and computational performance of the hybrid model and the MISO's prototype CFBM model in [10]. All test cases are performed on Intel Xeon X5680 @3.33 GHz RAM 32 GB on with AIMMS 4.33.2 [48] and CPLEX 12.7. The required relative MIP gap of the MISO system is 0.1%.

The following four formulations are compared with CFBM. Modeling techniques discussed in Sections III.C and D are applied to all the four hybrid models and CFBM.

- *HM-1:* Formulation of the hybrid model discussed in Section II, including constraints (8)–(13), (16)–(20), (23)–(25), and (28);
- *HM-2:* CFBM [10] with only min-up/down time constraints enforced on individual PTs instead of configurations;
- *HM-3:* HM-1 in which time-dependent startup indicators are declared as continuous;
- *HM-4:* The model by adding (27) in HM-1.

Table XI summarizes computational performance of HM-1 to HM-4 against CFBM, and detailed test results are shown in Table XII. As indicated, the hybrid model can provide significant cost savings compared with CFBM, with an average around $10,000. Some cases present the same CCGT scheduling with simliar costs (i.e., S2, S6, and S9), because individual PTs constraints are not binding. It is noteworthy that negative cost is mainly from discrepancy in MIP gap when terminated. While for other cases with minimum up/down time constraints binding in CFBM, the hybrid model shows great benefit with more

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TABLE XII
TEST RESULTS ON THE MISO SYSTEM

	Model	Obj ($)	Saving ($)	Gap (%)	R (s)	Time (s)
S1	CFBM	52441054	N/A	0.035	191	465
	HM-1	52409022	32032	0.080	187	478
	HM-2	52421267	19787	0.075	176	485
	HM-3	52415287	25767	0.098	169	536
	HM-4	52406778	34276	0.062	177	465
S2	CFBM	76581171	N/A	0.039	205	461
	HM-1	76585759	-4588	0.096	193	356
	HM-2	76567416	13755	0.054	226	444
	HM-3	76593555	-12384	0.099	193	355
	HM-4	76574754	6417	0.070	201	425
S3	CFBM	49313006	N/A	0.07	208	615
	HM-1	49291085	21921	0.094	200	599
	HM-2	49270622	42384	0.047	223	700
	HM-3	49289208	23798	0.089	192	694
	HM-4	49294244	18762	0.099	218	514
S4	CFBM	23450953	N/A	0.097	367	967
	HM-1	23445556	5397	0.063	407	934
	HM-2	23447644	3309	0.085	386	782
	HM-3	23447961	2992	0.094	347	984
	HM-4	23443622	7331	0.088	431	857
S5	CFBM	29935140	N/A	0.054	350	830
	HM-1	29922179	12961	0.085	465	1038
	HM-2	29926179	8961	0.095	363	583
	HM-3	29925383	9757	0.076	278	1154
	HM-4	29922179	12961	0.09	435	790
S6	CFBM	42749596	N/A	0.083	538	837
	HM-1	42753672	-6926	0.087	396	903
	HM-2	42743658	5938	0.07	465	734
	HM-3	42756487	-6891	0.089	387	849
	HM-4	42739655	9941	0.074	468	788
S7	CFBM	44873467	N/A	0.093	210	671
	HM-1	44858988	14479	0.049	308	661
	HM-2	44868110	5357	0.094	291	506
	HM-3	44865415	8052	0.099	285	704
	HM-4	44861368	12099	0.092	311	517
S8	CFBM	25112721	N/A	0.088	369	959
	HM-1	25103264	9457	0.056	360	929
	HM-2	25107710	5011	0.077	400	774
	HM-3	25104717	8004	0.064	354	947
	HM-4	25103708	9013	0.099	439	808
S9	CFBM	41371775	N/A	0.090	200	374
	HM-1	41372743	-281	0.093	193	338
	HM-2	41372056	-500	0.077	206	396
	HM-3	41372275	3718	0.075	198	408
	HM-4	41368057	-968	0.099	275	358
S10	CFBM	34973430	N/A	0.060	177	342
	HM-1	34967953	5477	0.056	193	338
	HM-2	34970081	3349	0.055	168	258
	HM-3	34971289	2141	0.088	181	462
	HM-4	34963514	9916	0.099	185	259

transitions observed in commitment schedule and lower total operation cost, which proves the enhanced flexibility provided by the hybrid model. Valuing certain physical features from individual components instead of aggregated features from configurations in CFBM, the hybrid model could also exploit fast ramping benefit of CCGTs more effectively, for handling various uncertainties from deeper penetrated renewable resources.

In terms of computational effort, with a relative optimality MIP gap of 0.1%, HM-1 shows very close performance as CFBM for both root-node relaxation time and total time. Although the data issues of different models discussed in Section IV.A.2 should not be neglected, results indeed show that total computational times of the hybrid model and CFBM [10] would be at the same magnitude. With minimum up/down time enforced on individual PTs instead of configurations, HM-2 shows slightly better performance compared with CFBM and HM-1. In addition, although HM-3 with continuous startup indicator consumes less time in root-node relaxation, it may suffer slightly higher total computational time than CFBM. That is, it would be better to define them as binary variables, which is also consistent with the conclusion on binary/continuous choice of transition variables. Computational results between HM-1 and HM-4 show similar trends for the tradeoff between root node relaxation time and total computational time as observed in 118-bus system studies. That is, it helps reducing the total computational time by more than 10% compared with HM-1, although the root node relaxation time is increased due to the involvement of more constraints.

TABLE XIII
SOLUTION STATISTICS COMPARISON OF HM-1 AND HM-4 FOR MISO CASE S10 FROM CPLEX 12.7

	HM-1	HM-4
Best Bound ($)	3.90576×10^6	3.49287×10^7
# of Clique	114	122
# of Bound	507	562
# of Flow	151	215
# of MIR	337	554
# of Zero-half	5	24
# of Gomory	11	38

Table XIII shows solutions of HM1 and HM-4 for the MISO case S10 solved via CPLEX 12.7, including root-node relaxation values (best bound) and numbers of various cutting planes. In this case, cutting planes generated from CPLEX include clique cuts, implied bound cuts, flow cuts, mixed integer rounding cuts, zero half cuts, and Gomory fractional cuts. After adding constraints (27), the root-node relaxation value after the presolving process is dramatically improved and more cutting planes are induced, which all indicates a stronger lower bound and a tighter feasible region with the help of build-in techniques from solver. As the objective of LP relaxation after presolving is closer to the optimal integer solution ($\$3.496 \times 10^7$ in this case), it shows the beneficial impact of adding constraint (27) to computational performance. Future work could include exploring additional formulation techniques and tightening constraints to further improve performance of the hybrid model.

V. CONCLUSIONS

This paper proposes a hybrid CCGT model, by exploiting market offers from both configurations and individual PTs, for reflecting physical operation features more accurately and operating CCGTs more economic efficiently. The proposed hybrid model is compared with CFBM and EBM, as well as various tightening formulations are analyzed to further improve computational performance of the hybrid model. Numerical studies show promising benefits on achieving better social welfare with reasonable computational effort. The next step is to work close with MPs for additional practical needs and to further improve performance of the hybrid model with state-of-the-art solvers.

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APPENDIX
AN EXAMPLE TO ILLUSTRATE TIGHTNESS OF THE HYBRID MODEL WITH CONSTRAINTS (28)

Considering a CCGT with three valid configurations of all off, 1CT, and 1CT1ST, denoted as $\boldsymbol{u} = (u^1, u^2, u^3)$, respectively. The feasible transition is defined as $\boldsymbol{v} = (v^{12},\ v^{21},\ v^{23}, v^{32})$. Constraints (9) and (10) are expressed as:

$$u_t^1 - u_{t-1}^1 = v_t^{21} - v_t^{12}$$
$$u_t^2 - u_{t-1}^2 = \left(v_t^{12} + v_t^{32}\right) - \left(v_t^{21} + v_t^{23}\right)$$
$$u_t^3 - u_{t-1}^3 = v_t^{23} - v_t^{32}$$
$$u_t^1 + u_t^2 + u_t^3 = 1$$
$$u_{t-1}^1 + u_{t-1}^2 + u_{t-1}^3 = 1$$

With min-up/down time of CT and ST being 1 hour, min-up/down constraints (16) and (17) are further described as:

CT: $v_t^{12} \le u_t^2 + u_t^3;\ v_t^{21} \le 1 - (u_t^2 + u_t^3)$

ST: $v_t^{23} \le u_t^3;\ v_t^{32} \le 1 - u_t^3$

It could be easily verified that when initial hour condition $\boldsymbol{u}_{t-1} = (1, 0, 0)$, the fractional solution $\boldsymbol{u}_t = (0, 0.5, 0.5)$ and $\boldsymbol{v}_t = (1, 0, 0.5, 0)$ is feasible for the LP relaxation of the integrated polytope described by constraints (9), (10) and (16), (17). However, constraint (28) could help cut off this solution as $\boldsymbol{v}_t$ violates

$$v_t^{12} + v_t^{21} + v_t^{23} + v_t^{32} \le 1$$

That being said, constraint (28) presents a tighter feasible region for LP relaxation of the original MIP problem.

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